

2026 Spring

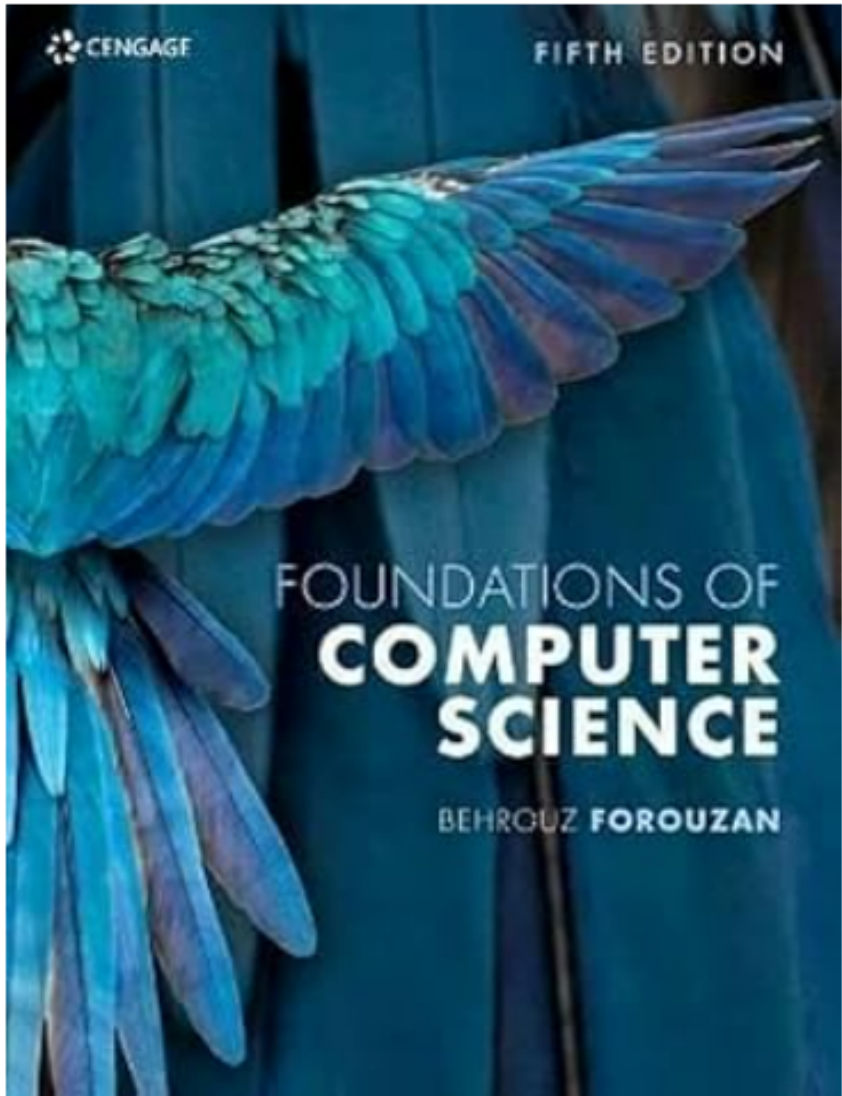
Course ID: 1142-1764

Foundations of Computer Science



13-Cloud Architecture

The Textbook



Foundations of Computer Science Paperback



作者 [Behrouz Forouzan](#) (Author)

The symposium was held Oct.-Nov. 1989, Research Triangle Park, North Carolina. One hundred papers in theoretical computer science treat dispersers, deterministic amplification, and weak random sources; efficient NC algorithms for set cover with applications to learning and geometry; the inverse of automorphism in polynomial time; and speeding-up linear programming using fast matrix multiplication. Acidic paper; no subject index. Annotation copyright Book News, Inc. Portland, Or.

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語言



English

尺寸



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Flip Classroom

Padlet Collaborative Learning

Introduction to Computer Science | Modules 5 – 10



M5 Networking



M6 Compute



M7 Storage



M8 Databases



M9 Architecture



M10 Auto Scaling



Your Group Builds the Knowledge Base — Together

Grading

The report's grading includes:

- 20% teacher's score
- 20% group score
- 60% peer evaluation.

Group leaders:

Module 6 - 411425209 LE TUAN ANH 黎俊英

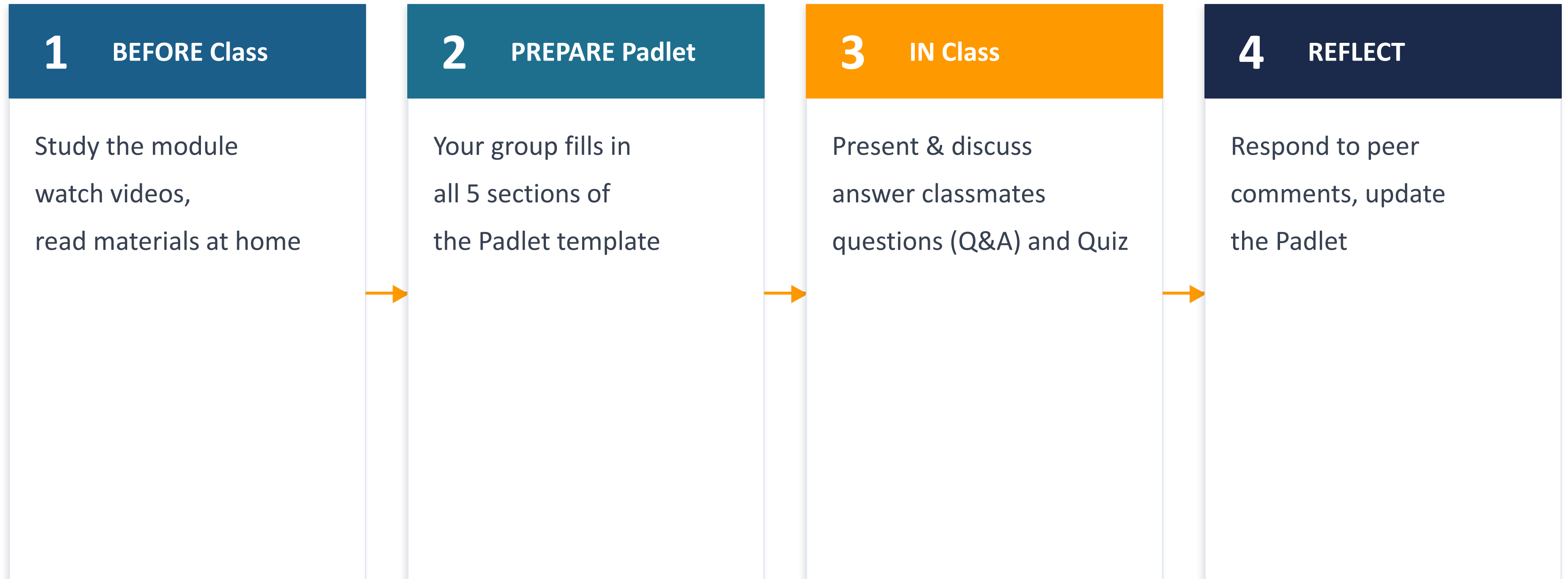
Module 7 - 411416878 CALVIN THEODORIC LAVENCIO 鄭鎧文

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Flip Classroom for Foundations of Computer Science



 Padlet is the shared workspace where your group documents your learning for each module — visible to the whole class!



Your Padlet Template — 5 Sections

S1



Learning Objectives

State the key learning goals of the module.
What should you understand by the end?

S2



Core Concepts

Summarize essential technical concepts,
terminology, and architectures.

S3



Labs

Document hands-on AWS lab steps,
screenshots, and observations.

S4



Challenges

Record difficult questions, errors
you encountered, and how you solved them.

S5



Q&A & Peer Support

Post questions for the class.
Answer and support your peers.



Tips for a Great Padlet

Add images & diagrams, not just text

Link to AWS documentation & tutorials

Use clear headings in each column

Respond to at least 2 peers in S5



MODULE 9

Cloud Architecture

AWS Academy Cloud Foundations

- ✓ AWS Well-Architected Framework
- ✓ Reliability & High Availability
- ✓ AWS Trusted Advisor

Module Overview

Learning Objectives

- Describe the AWS Well-Architected Framework and its five pillars
- Identify the design principles of each pillar
- Explain the importance of reliability and high availability
- Identify how AWS Trusted Advisor helps customers
- Interpret AWS Trusted Advisor recommendations

Topics & Activities

Topics:

- AWS Well-Architected Framework
- Reliability and High Availability
- AWS Trusted Advisor

Activities:

- Well-Architected Framework Design Principles (activity)
- Interpret AWS Trusted Advisor Recommendations
- Knowledge Check

AWS Well-Architected Framework

A guide to building secure, high-performing, resilient, and efficient infrastructure for cloud applications and workloads.



Operational Excellence

Run & monitor systems
to deliver business value



Security

Protect information,
systems, and assets



Reliability

Prevent & recover from
failures



Performance Efficiency

Use resources efficiently
as demands change



Cost Optimization

Run systems at the
lowest price point



Pillar 1 — Operational Excellence

Focus: Run and monitor systems to deliver business value, and to continually improve supporting processes and procedures.

Design Principles

- Perform operations as code
- Annotate documentation
- Make frequent, small, reversible changes
- Refine operations procedures frequently
- Anticipate failure
- Learn from all operational events and failures

Key Topics

Prepare

- Determine priorities
- Design workload for observability
- Reduce defects, ease remediation
- Mitigate deployment risks
- Know your readiness to support workload

Operate

- Understand health of workload
- Understand health of operations
- Manage workload and operations events

Evolve



Pillar 2 — Security

Focus: Protect information, systems, and assets while delivering business value through risk assessments and mitigation strategies.

Design Principles

- Implement a strong identity foundation
- Enable traceability
- Apply security at all layers
- Automate security best practices
- Protect data in transit and at rest
- Keep people away from data
- Prepare for security events

Key Areas

Identity & Access Mgmt

- Manage credentials and authentication
- Control human and programmatic access

Detective Controls

- Detect and investigate security events
- Defend against emerging security threats

Infrastructure Protection

- Protect your networks
- Protect your compute resources

Data Protection

- Classify, protect data at rest and in transit

Incident Response



Pillar 3 — Reliability

Focus: Prevent and quickly recover from failures to meet business and customer demand.

Design Principles

- Test recovery procedures
- Automatically recover from failure
- Scale horizontally to increase aggregate availability
- Stop guessing capacity
- Manage change in automation

Key Questions

Foundations

- How do you manage service quotas?
- How do you manage your network topology?

Change Management

- How does your system adapt to changes in demand?
- How do you monitor your resources?
- How do you implement change?

Failure Management

- How do you back up data?
- How does your system withstand component failure?
- How do you plan for disaster recovery?



Pillar 4 — Performance Efficiency

Focus: Use IT and computing resources efficiently to meet system requirements and maintain efficiency as demand changes.

Design Principles

- Democratize advanced technologies
- Go global in minutes
- Use serverless architectures
- Experiment more often
- Have mechanical sympathy

Key Areas

Selection

- Compute, Storage, Database, Network solutions
- Best-performing architecture choices

Review

- Evolve workload for new releases
- Stay up to date with AWS service improvements

Monitoring

- Ensure resources perform as expected
- CloudWatch alarms and metrics

Tradeoffs

- Use tradeoffs to improve performance
- Balance consistency, durability, and speed



Pillar 5 — Cost Optimization

Focus: Run systems to deliver business value at the lowest price point.

Design Principles

- Adopt a consumption model
- Measure overall efficiency
- Stop spending money on data center operations
- Analyze and attribute expenditure
- Use managed/app-level services to reduce TCO

Key Questions

Expenditure Awareness

- Govern usage
- Monitor usage and cost
- Decommission resources

Cost-Effective Resources

- Evaluate cost when selecting services
- Use pricing models to reduce cost

Supply & Demand

- Match supply of resources with demand
- Right-size instances and services

Optimize Over Time

- Evaluate new services for savings

Reliability & Availability

Reliability

A measure of a system's ability to provide functionality when desired by the user. Includes all system components: hardware, firmware, and software.

MTTF	MTTR	MTBF
Mean Time To Failure	Mean Time To Repair	Mean Time Between Failures
Expected operating time before first/next failure	Time taken to restore system after a failure	MTBF = MTTF + MTTR Higher MTBF = more reliable

Availability

Normal operation time ÷ Total time
Expressed as a percentage of uptime over a defined period.

Availability	Downtime per Year	Downtime per Month
99%	3.65 days	7.2 hours
99.9%	8.7 hours	43.2 minutes
99.95%	4.4 hours	21.6 minutes
99.99%	52.6 minutes	4.3 minutes
99.999%	5.3 minutes	26 seconds

High Availability

A highly available system can withstand some measure of degradation while still remaining available. Downtime is minimized and minimal human intervention is required.



Fault Tolerance

The built-in redundancy of an application's components and its ability to remain operational even when some components fail.

Examples:

- Multi-AZ deployments
- Redundant EC2 instances
- Elastic Load Balancing



Scalability

The ability of an application to accommodate increases in capacity needs without changing its design.

Examples:

- Auto Scaling Groups
- Horizontal scaling (adding more instances)
- Vertical scaling (bigger instance sizes)



Recoverability

The processes, policies, and procedures related to restoring service after a catastrophic event.

Examples:

- AWS Backup / snapshots
- Route 53 DNS failover
- Cross-Region replication

AWS Trusted Advisor

An online tool providing real-time guidance to help you provision resources following AWS best practices. Looks at your entire AWS environment and gives real-time recommendations in five categories.



Cost Optimization

Identify unused resources, Reserved Instance opportunities, and ways to reduce your AWS bill.

- Idle EC2 instances
- Underutilized RDS
- Unassociated Elastic IPs



Performance

Improve speed and responsiveness of your applications by checking service limits and high-utilization instances.

- High-utilization EC2
- CloudFront optimizations
- Content delivery



Security

Improve security by closing gaps, enabling various features, and examining permissions.

- S3 bucket permissions
- MFA on root account
- Security groups



Fault Tolerance

Increase availability and redundancy of your AWS application by taking advantage of auto-scaling, health checks, and Multi-AZ.

- Auto Scaling
- EBS snapshots
- Multi-AZ RDS



Service Limits

Notify you when service usage is more than 80% of the limit. Helps you avoid disruptions due to service throttling.

- EC2 instance limits
- VPC limits
- IAM limits



The AWS Well-Architected Framework has 5 pillars: Operational Excellence, Security, Reliability, Performance Efficiency, and Cost Optimization.



Reliability = MTTF + MTTR (MTBF). Availability is normal operation time ÷ total time. Five 9s = 99.999% uptime.



High availability requires Fault Tolerance, Scalability, and Recoverability working together.



AWS Trusted Advisor provides real-time best-practice guidance across 5 categories: Cost, Performance, Security, Fault Tolerance, Service Limits.